Predicting Infertility

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Developing a version of a Published Paper

Published Paper

- Predictors of miscarriage: a matched case-control study by Jalal Poorolajal et al

Our Dataset

- Infert Dataset is a built-in R dataset



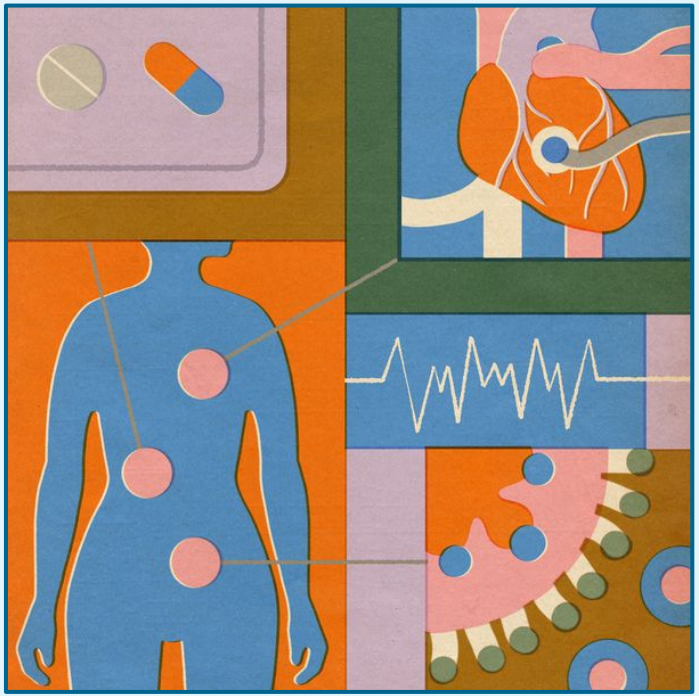




Why This Paper?

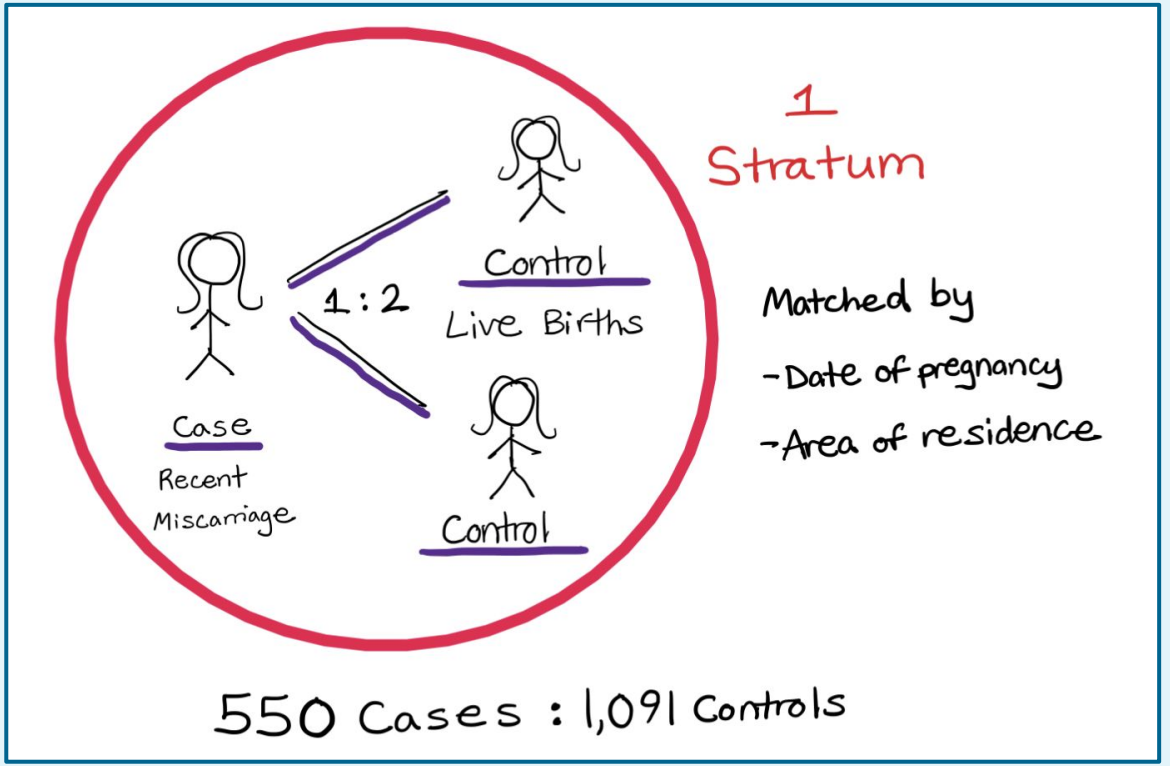
Women's Health

- Identifies major risk factors for miscarriage
- Guides prenatal care
- Paper uses data from Iran (low/middle-income country)



Published Paper: Predictors of Miscarriage

- Matched case-control
- Miscarriage vs Birth
- Multivariate Conditional Logistic Regression
- Odds ratios and 95% CI

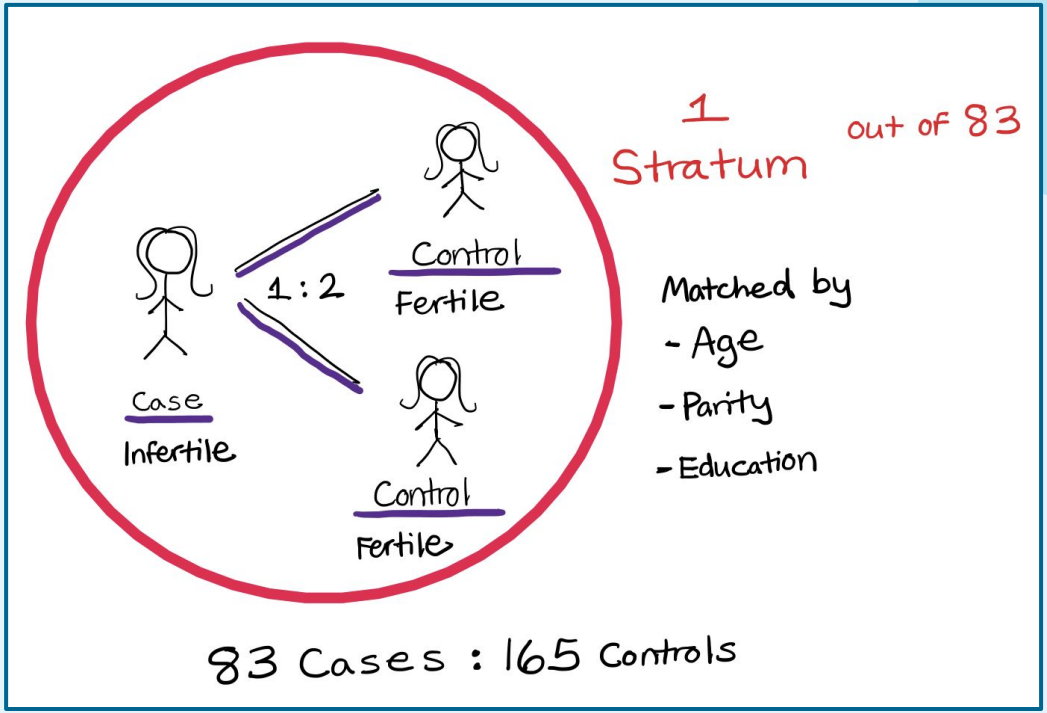


Our Dataset: Infert Dataset

Research Question:

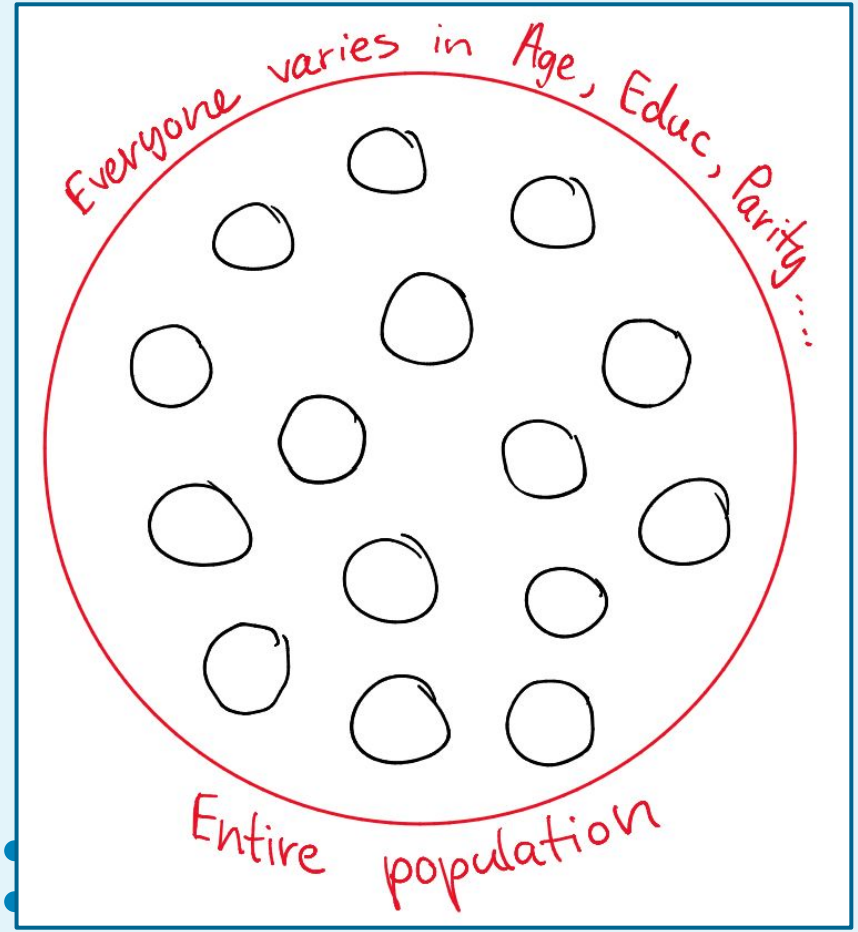
How do previous miscarriages and abortions affect infertility in women?

- Matched case control study (1: 2 ratio)
- Infertile vs Fertile women
- Induced vs spontaneous abortions



Types of Models





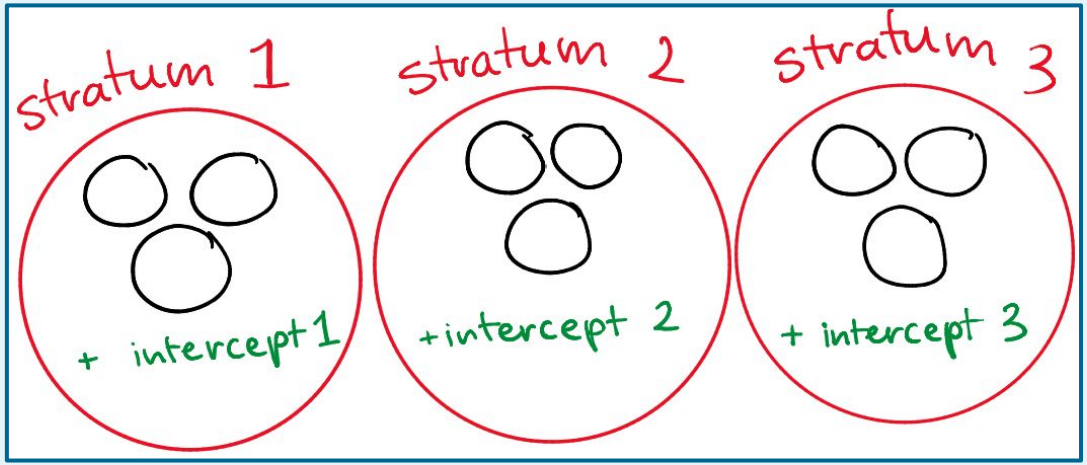
Ordinary Logistic Regression

- Completely ignores matched structure of data

- **Pros:**
  - Simple
  - Easy to Interpret
- **Cons:**
  - Ignores matching
  - Introduces confounding variables

```
Call:
glm(formula = case ~ age + parity + education + induced + spontaneous,
    data = infert)
```

Ordinary Logistic Regression + factor(stratum)

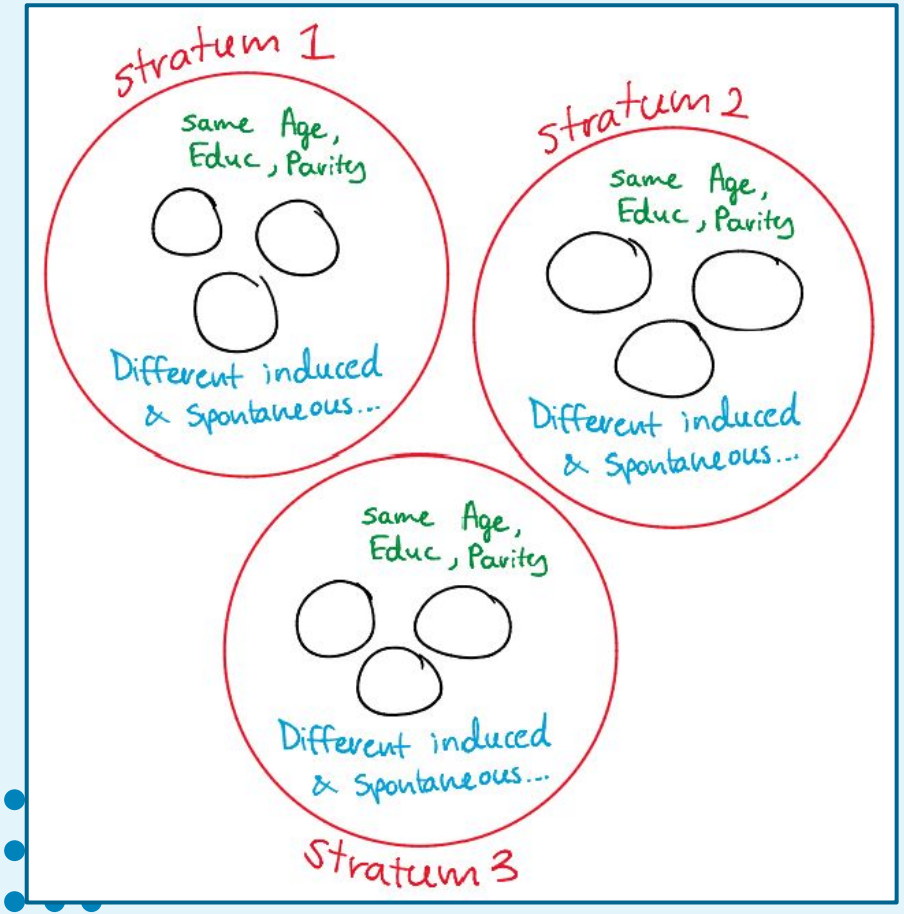


- Each stratum gets its own intercept (to attempt at matching)
- Few data points (3 women per) to work with, intercepts are estimated unstably

- **Pros:**
  - Addresses matching to some degree
- **Cons:**
  - Unstable intercepts
  - Untrustworthy interpretations

```
glm(formula = case ~ age + parity + education + induced + spontaneous +
    factor(stratum), data = infert)
```

Conditional Logistic Regression Model



- Compares within each matched set (considers matched structure of data)
- Sees true effects of abortions and miscarriages on infertility

- **Pros:**
  - Addresses Matching
  - Removes unnecessary intercepts
  - Less biased
- **Cons:**
  - None in the case of our research question

```
Call:
coxph(formula = Surv(rep(1, 248L), case) ~ induced + spontaneous +
    strata(stratum), data = infert, method = "exact")
```

Comparing These Models

AIC and BIC

Model Comparison Using AIC and BIC

Model	AIC	BIC
full_model_stratum	132.40	137.24
ord_reduced	286.05	307.13
ord_factored_reduced	299.93	313.99

Conditional Logistic Regression Model Interpretation

Answer Research Question:

How do previous miscarriages and abortions affect infertility in women?

- exp(coef) = odds ratio
- Induced Abortion → 4.1x greater odds of being infertile
- Spontaneous Abortion (miscarriage) → 7.3x greater odds of being infertile

```
Call:
coxph(formula = Surv(rep(1, 248L), case) ~ induced + spontaneous +
    strata(stratum), data = infert, method = "exact")

  n= 248, number of events= 83

               coef exp(coef) se(coef)     z Pr(>|z|)
induced     1.4090    4.0919   0.3607 3.906 9.38e-05 ***
spontaneous 1.9859    7.2854   0.3524 5.635 1.75e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

            exp(coef) exp(-coef) lower .95 upper .95
induced         4.092     0.2444     2.018     8.298
spontaneous     7.285     0.1373     3.651    14.536
```

Conditional Logistic Model Interpretation

- Induced 95% CI : (2.018 , 8.298)
- Spontaneous 95% CI : (3.651 , 14.536)
- Fairly wide but informative of the analysis and data setup

```
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```

Conclusion

How do previous abortions and miscarriages affect odds of infertility in women?

- Miscarriages cause higher risk of infertility than abortions
- Previous Abortion: 4.1x greater risk
- Previous Miscarriage: 7.3x greater risk

What is the usefulness of strata and conditional logistic regression?

- For stratified data Conditional Logistic Regression takes advantage by focusing on specific variables
- Estimates odds of being infertile within each matched set